GATE ALL BRANCHES



Probability and Statistics

-ENGINEERING MATHEMATICS

Lecture No.- 08



By- Rahul sir

Recap of Previous Lecture



Topic

Problems based on conditional probability and Bayes theorem

Topics to be Covered



Topic Problems based on Bayes theorem

Topic Concept of random variable

Topic Types of random variable



Topic : Probability and Statistics

#Q. Parcels from sender S to receiver R pass sequentially through two post – offices. Each post - office has a probability 1/5 of losing an incoming parcel, independently of all other parcels. Given that a parcel is lost, the probability that it was lost by the second post-office is.



Topic : Probability and Statistics

MSQ

$E_1, E_2, E_3 = \frac{1}{2}, \frac{1}{4}, \frac{1}{4}$

ship is operational at least Two engines

#Q. A ship is fitted with three engines E_1, E_2 and E_3. Then engines function independently of each other with respective 1/2,1/4,1/4. for the ship to be operational, at least two of its engines must function. Let X denote the event that the ship is operational and let X_1, X_2 and X_3 denote respectively the events that the engines E1, E2 and E3 and functioning. Which of the following is/are true?

A $P\left(\frac{X_1^c}{X}\right)=\frac{3}{16}$

B P (exactly two engines of the ship are functioning /X) = 7/8

C $P\left(\frac{X}{X_2}\right)=\frac{5}{16}$

D $P\left(\frac{X}{X_1}\right)=\frac{7}{16}$

Slide

$P(E_1)=\frac{1}{2}$ $P(E_2)=\frac{1}{4}$ $P(E_3)=\frac{1}{4}$

$P(X_1)=\frac{1}{2}$ $P(X_2)=\frac{1}{4}$ $P(X_3)=\frac{1}{4}$

If Two engines Are working Then Ship is operational

X_1 = engine E_1 is working

X_2 —— E_2

X_3 —— E_3

Ship is operational

X

E_1, E_2, E_3



$= E_1 \wedge E_2 \wedge E_3$ ✓

$= E_1 \wedge E_2 \wedge \bar{E}_3$ ✓

$= E_1 \wedge \bar{E}_2 \wedge E_3$ ✓

$= E_1 \wedge \bar{E}_2 \wedge \bar{E}_3$ X

$= \bar{E}_1 \wedge E_2 \wedge E_3$ ✓

$= \bar{E}_1 \wedge E_2 \wedge \bar{E}_3$ X

$= \bar{E}_1 \wedge \bar{E}_2 \wedge E_3$ X

$= \bar{E}_1 \wedge \bar{E}_2 \wedge \bar{E}_3$ X

Slide

X_1, X_2, X_3

Independent

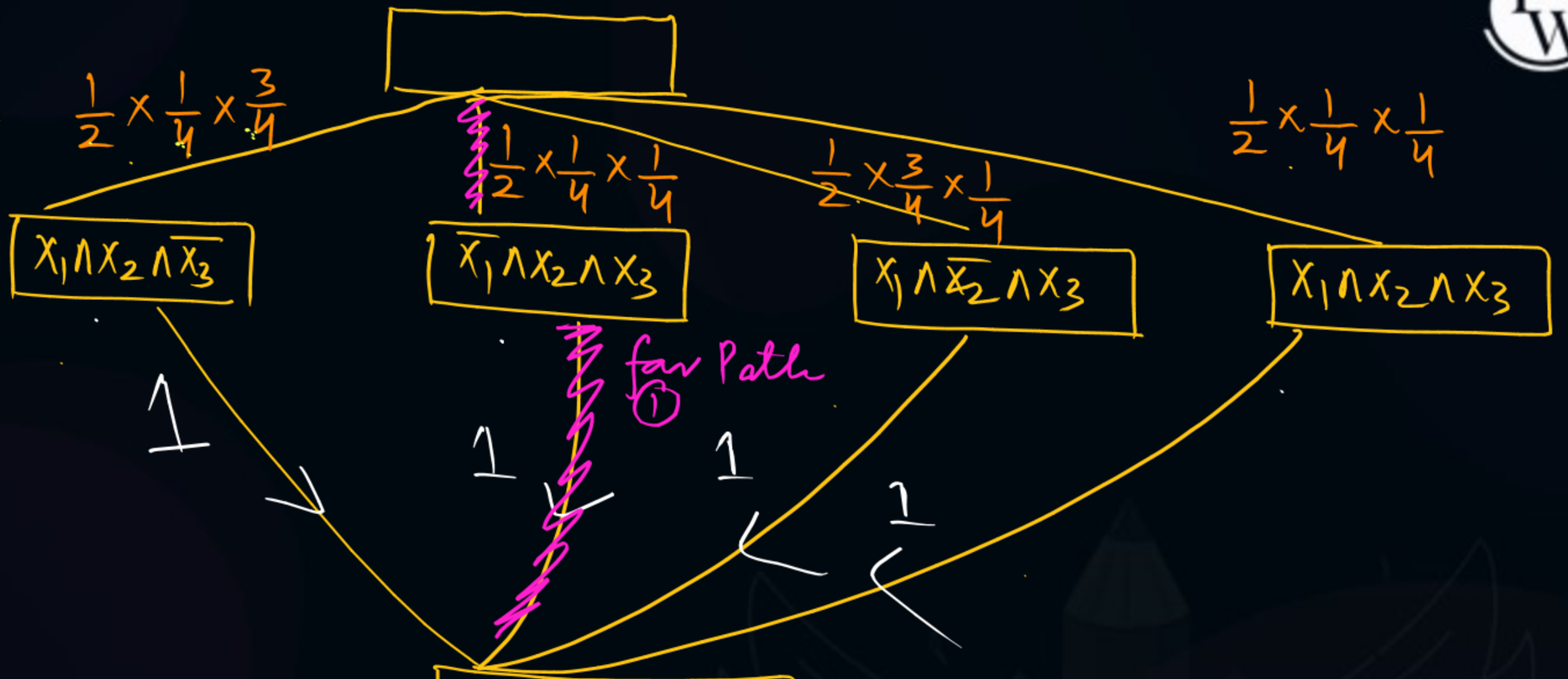
$P(X_1 \cap X_2 \cap X_3)$

$= P(X_1)\,P(X_2)\,P(X_3)$

$P(X_1) = \frac{1}{2}$

$P(X_2) = \frac{1}{4}$

$P(X_3) = \frac{1}{4}$



$$P\left(\frac{X_1^c}{X}\right) = P\left[\frac{X_1 \text{ is Not working}}{\text{Ship is operational}}\right]$$

$$\Rightarrow \frac{\frac{1}{2} \times \frac{1}{4} \times \frac{1}{4}}{\frac{3}{32} + \frac{1}{32} + \frac{3}{32} + \frac{1}{32}} = \frac{1}{8}$$

b) $P\left(\frac{\text{exactly Two engines}}{\text{ship is operational}}\right) = \frac{① + ② + ③}{① + ② + ③ + ④}$

$= \frac{7}{8}$ Ans (B)

c) $P\left(\frac{X}{X_2}\right)$ = Using conditional prob

$= \frac{P(X \cap X_2)}{P(X_2)} = \frac{P(\text{ship is operational} \cap X_2 \text{ is working})}{P(X_2)}$

$= \frac{① + ② + ④}{\frac{1}{4}} = \frac{\frac{1}{32} + \frac{3}{32} + \frac{1}{32}}{\frac{1}{4}}$

$= \frac{\frac{5}{32}}{\frac{1}{4}} = \frac{20}{32} = \frac{16}{16} = \frac{5}{8}$

$$P\left(\frac{X}{X_1}\right) = \frac{P(X \cap X_1)}{P(X_1)} = \frac{\frac{3}{32}+\frac{3}{32}+\frac{1}{32}}{\frac{1}{2}}$$

Using conditional formula

$$= \frac{\frac{7}{32}}{\frac{1}{2}} = \frac{7 \times 2}{32} = \frac{7}{16}$$

D

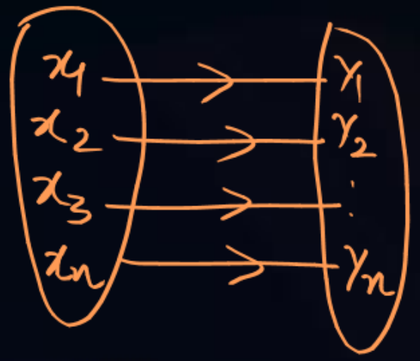
B D

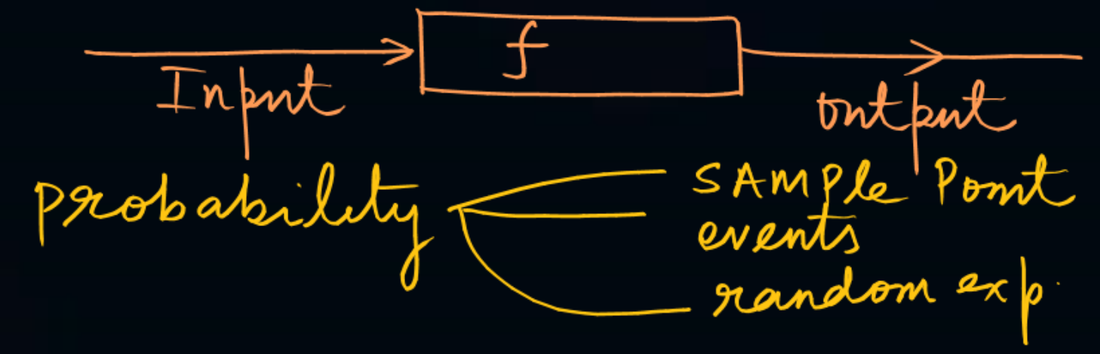
Random variable :

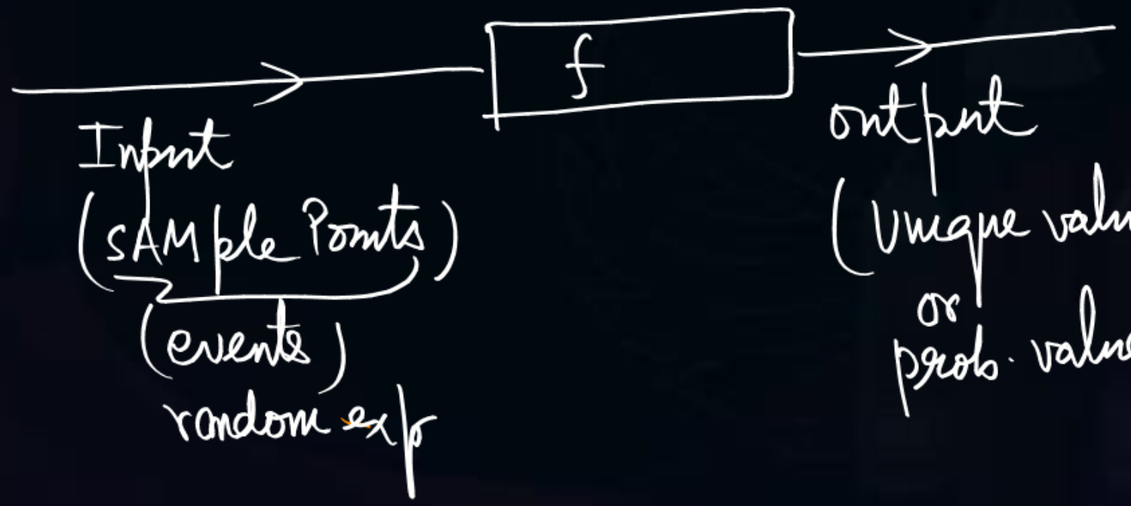
Random variable is a mathematical Function

$Y = f(x)$

$\{x = \text{Input}\ ]$
$[y = \text{output}\}$







Slide

Tossing A Two Coin

Re 1, Re 2

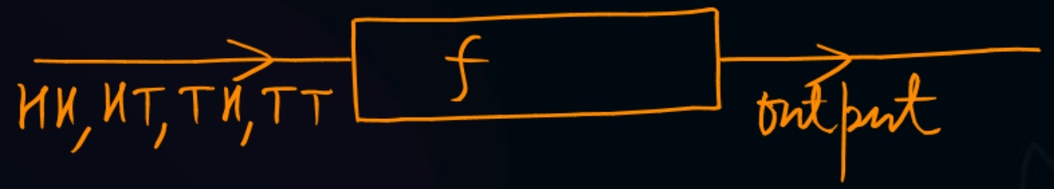
H < H = HH
T = HT
T < H = TH
T = TT

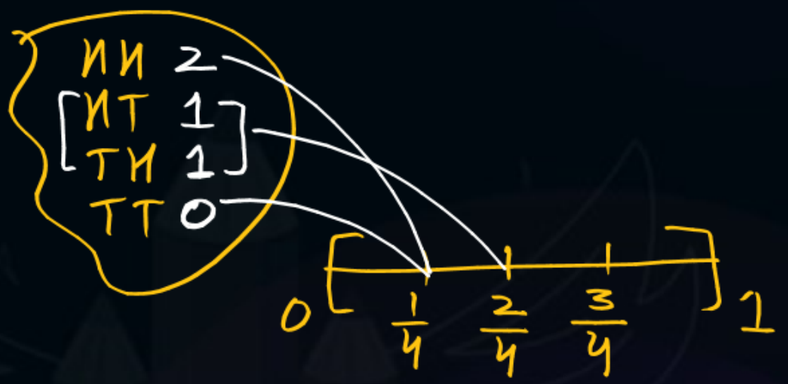
SAMPLE Point (SAMPLE space)

X = No. of HEADS / No. of tails

SAMPLE Point	HH	HT	TH	TT
X	2	1	1	0

X = 0, 1, 2







Integer value

output prob.

$0 \le P(E) \le 1$

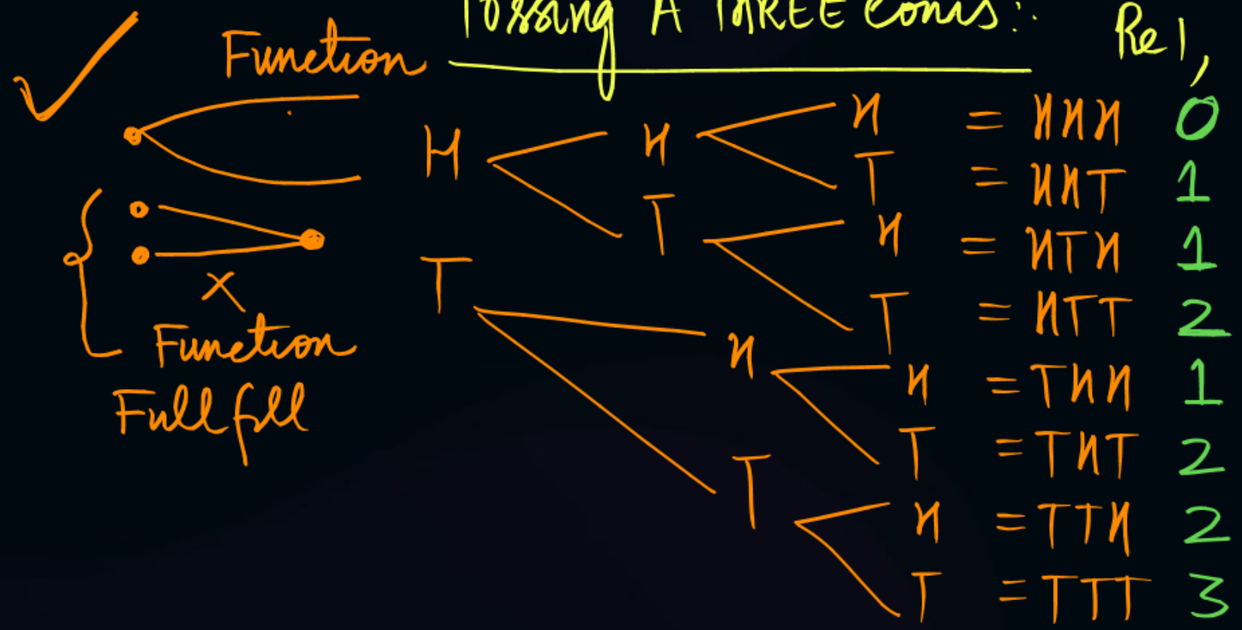
$P(X=0) = P(X=0\text{HEAD}) = \frac{1}{4}$

$P(X=1) = P(X=1\text{HEAD}) = \frac{2}{4} = \frac{1}{2}$

$P(X=2) = \frac{1}{4}$

Tossing A THREE coins :

Re1, Re2, Re5



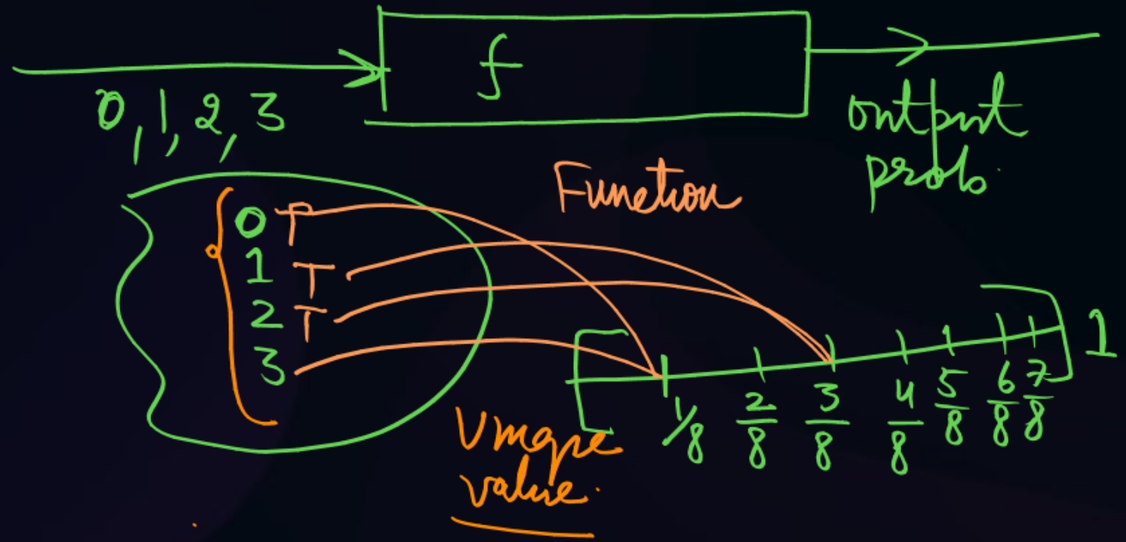
X = No. of Tails
(random variable)

$X = 0, 1, 2, 3$

" Random variable is a mathematical Function

$$f : X_S \longrightarrow P$$

$X_S \rightarrow 0, 1, 2, 3$



X	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

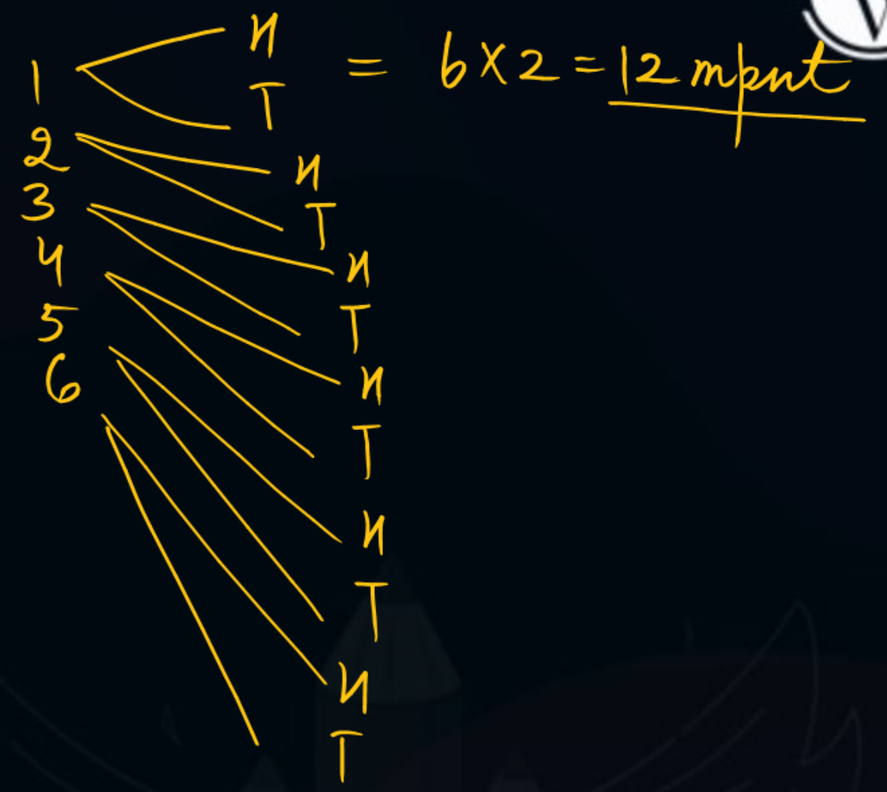
One Die and one Coin

Die → 1, 2, 3, 4, 5, 6

Coin → H, T

Independent

H1	T1
H2	T2
H3	T3
H4	T4
H5	T5
H6	T6



$= 6 \times 2 = 12$ mpnt

H1	H2	H3	T1	T2	T3	H4	H5	H6	T4	T5	T6
$\frac{1}{2}\times\frac{1}{6}$	$\frac{1}{2}\times\frac{1}{6}$	$\frac{1}{2}\times\frac{1}{6}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{12}$

Types of Random Variable:

1) Discrete Random variable
2) continuous Random variable

$f : X_S \longrightarrow [0,1]$ real valued

$\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^3 + \cdots \infty$

Countable Infinite

SUM = finite

Random variable

Discrete

Countable | Infinite

Add

- ✓ Tossing A coin
- ✓ Throwing A Dice
- ✓ Picking A ball
- ✓ Playing cards

X = Integer value

$X = 0, 1, 2, 3 \ldots$

(Integer)

Continuous

Uncountable

Infinite Pts

Infinite (Uncountable) = Interval

(Uncountable Infinite No)

- ✓ Height of a Person
- ✓ Discharging A Battery

$1 < X < 2$

$0 < X < 1$ Hrs

Infinite uncountable No.

Slide

Random variable

Discrete

(Arrival) Pattern

$X = 0, 1, 2, 3,$

(Integer value)

$(0, 1, 2, 3, 4, 5, 6)$

- ✓ Tossing A coin
- ✓ Picking A ball
- ✓ Throwing A Die

Continuous

(Waiting Time)

Integration

(Interval)

$$\int_A^B f(x)\,dx$$ ← function given



THANK - YOU